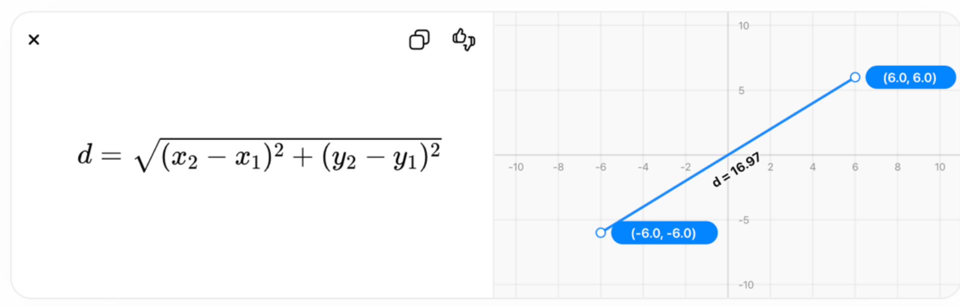
Euclidean Distance Calculation Example

Distance between two cities is the straight-line distance (like a ruler), computed using the Pythagorean theorem. An example on the first two cities in berlin52.tsp instance from the TSPLIB data (the Library of Traveling Salesman problems) is explained below:

- **The formula used:**



- **The two cities and their coordinates:**

City 1: $(x_1, y_1) = (565,575)$

City 2: $(x_2, y_2) = (25,185)$

- **Apply the formula to the two cities:**

1. **Compute differences**

$x_2 - x_1 = 25 - 565 = -540$
$y_2 - y_1 = 185 - 575 = -390$

2. **Square them**

$(-540)^2 = 291600$
$(-390)^2 = 152100$

3. **Add them**

$291600 + 152100 = 443700$

4. **Take square root**

$\sqrt{443700} \approx 666.1080993352356$

5. **Final result**

$d \approx 666.1080993352356$